

A Universal Exact Determinant Formula for Finite Subset-Action Atom Moduli

Integer cycle indices, sparse optimal circuits and a binomial-product rank certificate

mxym/math research project (AI-assisted)

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Abstract

For the natural action of S_n on k -subsets, constrain a probability law ν to give a uniform image for every prescribed k -subset. We obtain, for **every finite pair** $n \geq 1$, $0 \leq k \leq n$, an exact rational formula for the smallest constant comparing the probability change of any permutation atom with total variation. The formula uses only explicitly prescribed integer cycle-index coefficients, determinants of size at most $\min(k, n - k) + 1$, and a finite maximum. A closed product evaluates a distinguished full-rank determinant; an extremal-circuit lemma proves that at most $\min(k, n - k) + 2$ conjugacy classes suffice for exact attainment. Unlike the earlier orbital linear program, the answer has no continuous optimization variables, implicit infimum or solver dependency. It is an **explicit finite combinatorial closed formula**, not an elementary piecewise rational expression free of any finite maximum.

1. Problem and boundary cases

Let $\Omega_{n,k} = \binom{[n]}{k}$ and let u_n be uniform on S_n . A law ν on S_n is admissible when

$$\nu\{\pi : \pi(E) = F\} = \binom{n}{k}^{-1} \quad (E, F \in \Omega_{n,k}).$$

Set

$$C_{n,k} = \sup_{\substack{\nu \neq u_n \\ \nu \text{ admissible}}} \frac{|\nu(\text{id}) - 1/n!|}{\frac{1}{2} \sum_{\pi \in S_n} |\nu(\pi) - 1/n!|},$$

with supremum zero if the admissible set contains only the uniform law. By left translation, the same constant applies to every other specified permutation atom. Complementation of subsets is equivariant, so $C_{n,k} = C_{n,n-k}$.

Put $m = \min(k, n - k)$, always satisfying $n \geq 2m$. For the singleton group S_1 the constant is zero. For $n \geq 2$, $C_{n,0} = C_{n,n} = 1$: both actions are trivial and a signed identity/nonidentity perturbation attains one. For the remainder assume $n \geq 2$ and $1 \leq m \leq n/2$.

2. All integer data from (n, m)

The feasible short-cycle vectors are

$$\mathcal{T}_{n,m} = \left\{ c = (c_1, \dots, c_m) \in \mathbb{Z}_{\geq 0}^m : r_c = n - \sum_{j=1}^m j c_j \in \{0\} \cup \{m+1, m+2, \dots\} \right\}.$$

The canonical representative of c has c_j cycles of length j , together with one r_c -cycle if $r_c > 0$. The identity type $c_* = (n, 0, \dots, 0)$ is unique.

Define the two-state matrix and cycle-index polynomial recurrence

$$M(u, t) = \begin{pmatrix} 1 & u \\ 1 & ut \end{pmatrix}, \quad Z_0 = 2, \quad Z_1 = 1 + ut, \quad Z_\ell = (1 + ut)Z_{\ell-1} - u(t-1)Z_{\ell-2}.$$

Define $\Lambda(u, t) = \sum_{a \geq 0} A_a(t) u^a$ as the unique formal-series solution with constant term one of

$$\Lambda^2 - (1 + ut)\Lambda + u(t-1) = 0.$$

Its coefficients are integer polynomials, given by $A_0 = 1$ and

$$A_a = tA_{a-1} - \sum_{j=1}^{a-1} A_j A_{a-j} - \mathbf{1}_{\{a=1\}}(t-1).$$

For $c \in \mathcal{T}_{n,m}$, define **integers**

$$F_h(c) = [u^m t^h] \Lambda(u, t)^{r_c} \prod_{\ell=1}^m Z_\ell(u, t)^{c_\ell}, \quad 0 \leq h \leq m.$$

The exact all-rank cycle-index theorem (proved independently in Section 25 of the main dossier) shows that this counts the number of m -subsets E satisfying $|E \cap g(E)| = h$, for any g represented by c . In particular $F_h(c) \geq 0$ and $\sum_{h=0}^m F_h(c) = N := \binom{n}{m}$.

Define the $(m+1) \times |\mathcal{T}_{n,m}|$ **integer matrix**

$$A(n, m) = (a_c)_{c \in \mathcal{T}_{n,m}}, \quad a_c = (1, F_0(c), \dots, F_{m-1}(c))^T.$$

The first row encodes signed total mass zero. The other m rows encode all necessary independent marginal-orbital constraints; F_m is redundant by the fixed sum N .

3. Main exact formula

Order the nonidentity types in $\mathcal{T}_{n,m} \setminus \{c_*\}$ lexicographically. For each selection of $m+1$ **different** such types

$$T = \{c_1 < \cdots < c_{m+1}\},$$

form $B_T = (a_{c_*}, a_{c_1}, \dots, a_{c_{m+1}})$, an integer matrix of shape $(m+1) \times (m+2)$. For $i = 0, \dots, m+1$, put

$$\Delta_i(T) = (-1)^i \det(B_T \text{ with column } i \text{ deleted}).$$

Omit a selection when all $\Delta_i(T)$ vanish. An empty maximum means zero.

Theorem A (universal finite exact-value formula). For every $n \geq 2$ and $1 \leq m = \min(k, n-k) \leq n/2$,

$$C_{n,k} = \max_{\substack{T \subseteq \mathcal{T}_{n,m} \setminus \{c_*\} \\ |T|=m+1, \sum_{i=0}^{m+1} |\Delta_i(T)| > 0}} \frac{2|\Delta_0(T)|}{\sum_{i=0}^{m+1} |\Delta_i(T)|}.$$

Together with the boundary cases in Section 1, this is a **terminating exact rational arithmetic formula valid for every finite** (n, k) . If $C_{n,k} > 0$, an attaining marginal-preserving signed perturbation exists with support on at most $m+2$ conjugacy classes. It is converted into a genuine nonnegative probability law for all sufficiently small positive rational amplitudes.

The proof consists of the following independent nonzero determinant evaluation and a finite-dimensional circuit lemma.

4. The exact full-rank determinant

Theorem B (sharp rank theorem). For $n \geq 2m \geq 2$,

$$\text{rank}_{\mathbb{Q}} A(n, m) = m+1.$$

For $j = 0, \dots, m-1$, let g_j have exactly j fixed points and one long cycle of length $n-j$. Then

$$\det(a_{g_0}, \dots, a_{g_{m-1}}, a_{c_*}) = (-1)^m \prod_{r=0}^{m-1} \binom{n-2r}{m-r} \neq 0.$$

Proof. For each $j < m$, its only nontrivial cycle has length $n-j \geq m+1$. It fixes no m -subset, because the j fixed points are fewer than m . Thus its orbital generating polynomial

$$G_j(t) = \sum_{h=0}^m F_h(g_j) t^h = [u^m] (1+ut)^j \Lambda(u, t)^{n-j}$$

has degree at most $m-1$. Let ∇ denote forward differences in j . The binomial theorem gives

$$\nabla^r G_0(t) = [u^m] \Lambda(u, t)^{n-r} (1+ut - \Lambda(u, t))^r.$$

Differentiating the defining quadratic equation for Λ gives the exact series identities

$$\Lambda(u, 1) = 1+u, \quad \partial_t \Lambda(u, 1) = \frac{u^2}{1+u}.$$

Consequently $1 + ut - \Lambda(u, t)$ vanishes at $t = 1$ and has first derivative $u/(1 + u)$. Therefore the polynomial $\nabla^r G_0$ vanishes to order at least r at $t = 1$, with **exact diagonal Taylor coefficient**

$$\frac{1}{r!}(\nabla^r G_0)^{(r)}(1) = [u^m](1 + u)^{n-r} \left(\frac{u}{1 + u} \right)^r = \binom{n-2r}{m-r} > 0 \quad (0 \leq r < m).$$

Changing the column basis from $G_0, \dots, G_{m-1}$ to their successive forward differences is a unit-triangular transformation. Changing the row basis from coefficients of $t^0, \dots, t^{m-1}$ to Taylor coefficients at $t = 1$ is also unit triangular. In these bases the resulting matrix is triangular with diagonal (14). Thus

$$\det(F_h(g_j))_{0 \leq h, j < m} = \prod_{r=0}^{m-1} \binom{n-2r}{m-r}.$$

The identity column is $(1, 0, \dots, 0)^T$, so expansion along this last column yields the sign $(-1)^m$, proving (10) and (9). QED.

This is a **closed product identity for all** n, m , not a finite pattern inferred from numerical determinant evaluations.

5. A circuit theorem for signed ℓ^1 kernels

Lemma C (maximal-minor circuit formula). Let $D = (d_0, \dots, d_{p-1})$ be a real $r \times p$ matrix of full row rank r , whose first row consists entirely of ones. Put

$$\Gamma(D) = \sup_{\substack{v \neq 0 \\ Dv=0}} \frac{2|v_0|}{\sum_i |v_i|},$$

and define it to be zero if $\ker D = \{0\}$. For each r -subset $J \subseteq \{1, \dots, p-1\}$, form the $r \times (r+1)$ matrix $(d_0, d_{j_1}, \dots, d_{j_r})$ and its alternating maximal-minor vector $\Delta(J)$ as in (7). Then

$$\Gamma(D) = \max_{\substack{J \subseteq \{1, \dots, p-1\} \\ |J|=r, \sum_i |\Delta_i(J)| > 0}} \frac{2|\Delta_0(J)|}{\sum_i |\Delta_i(J)|}.$$

An extremizer exists with support on at most $r+1$ columns.

Proof. Every alternating maximal-minor vector belongs to the kernel of its rectangular matrix, by determinant expansion along a duplicated row. Extending it by zeros shows that every ratio on the right of (15) is bounded above by $\Gamma(D)$.

Conversely suppose $\Gamma(D) > 0$. Compactness of the intersection $\ker D \cap \{\sum |v_i| = 2\}$ yields an extremizer v . Reverse its sign if necessary so $v_0 > 0$; normalize $\sum_{v_i > 0} v_i = 1$. Since the first row of D is all ones, the negative mass also equals one. Fix the sign orthant of v . The corresponding nonnegative-coordinate feasible set

$$\mathcal{P} = \{x \geq 0 : D \operatorname{diag}(\varepsilon)x = 0, \sum_{\varepsilon_i = +1} x_i = 1\}$$

is a compact polytope, as both positive and negative masses equal one. Choose a vertex of $\mathcal{P}$ maximizing its identity coordinate, which stays strictly positive.

Let S be that vertex's nonzero support and $w \in \ker D_S$ its signed coordinate vector, all of whose S entries are nonzero. If $\dim \ker D_S \geq 2$, there exists a nonzero kernel direction preserving the one scalar positive-mass normalization. Sufficiently small positive and negative multiples of this direction keep all

support signs unchanged and produce distinct points of $\mathcal{P}$ with the original vertex as their midpoint. This contradicts extremality. Therefore

$$\dim \ker D_S = 1, \quad \text{rank } D_S = |S| - 1.$$

Deleting any one column of S leaves an independent family: otherwise there would be a second dependence not proportional to w , whose deleted coordinate is nonzero. Hence S is an **elementary circuit** and $|S| \leq r + 1$. It contains column zero because $v_0 > 0$.

Extend the independent family $S \setminus \{0\}$ to a basis of the full row space using columns from the entire matrix D . Adjoin column zero, obtaining exactly $r + 1$ columns of rank r . Its kernel is one dimensional and contains the original circuit vector with zeros in the added coordinates. Thus it is proportional to the alternating maximal-minor vector of this $r \times (r + 1)$ matrix, and its norm ratio appears among the right-hand terms of (15). This establishes the reverse inequality. When $\Gamma(D) = 0$, the first direction forces every candidate ratio to vanish, so the statement also holds. QED.

6. Completion of the universal formula and attainment

We recall precisely why orbital compression does not lose any signed marginal-preserving perturbation. Conjugation averaging of a signed law preserves its identity atom and image-marginal constraints and cannot increase total variation. For a conjugation-invariant law, equality of all image marginals is equivalent to equality of the aggregate orbital moments $F_0, \dots, F_m$. The total-mass row and the first m moments suffice because $\sum F_h = N$.

Classes with identical $c = (c_1, \dots, c_m)$ have identical moments by (5), so their signed masses may be aggregated, never increasing the ℓ^1 mass. Conversely, every short-cycle vector has a canonical actual conjugacy-class representative. Any real vector $v = (v_c) \in \ker A(n, m)$ with total mass zero can be represented by placing its signed class mass uniformly on these canonical classes. For sufficiently small positive amplitude ε , adding εv to the strictly positive uniform law gives a **nonnegative admissible probability distribution**, with identity atom change εv_{c_*} and total variation $(\varepsilon/2) \sum_c |v_c|$. Therefore

$$C_{n,k} = \Gamma(A(n, m)).$$

Theorem B gives $\text{rank } A = m + 1$. Lemma C with $r = m + 1$ yields exactly Theorem A, equation (8).

For explicit attainment orient a maximizing minor vector so $\Delta_0 > 0$ and write

$$d = \sum_{\Delta_i > 0} \Delta_i = - \sum_{\Delta_i < 0} \Delta_i > 0.$$

Define central probability laws supported on the selected canonical classes by

$$P = \sum_{\Delta_i > 0} \frac{\Delta_i}{d} U_{[g_i]}, \quad Q = \sum_{\Delta_i < 0} \frac{-\Delta_i}{d} U_{[g_i]}.$$

Their supports are disjoint; since $A\Delta = 0$, their image marginals agree **exactly**. For small enough rational $\varepsilon > 0$, $\nu_\varepsilon = u_n + \varepsilon(P - Q)$ is nonnegative, admissible and has total variation ε . Its identity atom excess is $\varepsilon\Delta_0/d$, giving

$$\frac{\nu_\varepsilon(\text{id}) - 1/n!}{\|\nu_\varepsilon - u_n\|_{\text{TV}}} = \frac{2|\Delta_0|}{\sum_i |\Delta_i|} = C_{n,k}.$$

Thus Theorem A provides both an exact upper value and a matching explicit probability construction. QED.

7. Complexity, rational height, and examples

The number $p = |\mathcal{T}_{n,m}|$ obeys

$$p \leq \prod_{j=1}^m (1 + \lfloor n/j \rfloor) = O_m(n^m).$$

There are at most $\binom{p-1}{m+1} = O_m(n^{m(m+1)})$ ratios in (8). Each uses $m+2$ integer determinants of size $m+1$, evaluable by fraction-free Bareiss elimination. Hence for every **fixed** m the formula yields a polynomial-in- n terminating integer algorithm; it is not claimed polynomial in the binary input length $\log n$.

Because $0 \leq F_h(c) \leq N = \binom{n}{m}$, cofactor expansion along the row of ones gives

$$|\Delta_i| \leq (m+1)! N^m.$$

The first row also gives $\sum_i \Delta_i = 0$. Thus every resulting reduced rational optimum has denominator at most

$$(m+2)(m+1)! \binom{n}{m}^m.$$

The independent pure-integer implementation uses the published all-rank cycle recurrence to compute F_h , fraction-free Bareiss integer determinants to compute every Δ_i , and Python Fraction only for exact quotient comparison. Its fixed regression set reproduces, among others,

(n, k)	(3, 1)	(4, 2)	(6, 3)	(8, 3)	(9, 3)	(10, 3)	(8, 4)	(9, 4)
$C_{n,k}$	1/3	1/3	5/14	89/244	259/691	368/935	5/14	5/14.

It checks the exact rank-product identity (10) for $m \leq 9$ and multiple degrees. These finite tests are supplemental; the complete proof applies to every finite n, k .

8. Scope and provenance

This theorem completely answers arbitrary finite-parameter **exact evaluability** in a prescribed determinant/minor sense. It does **not** produce a single short algebraic rational function or a finite number of parameter branches valid for all (n, k) ; the finite maximization may be very large. A uniform classification of which circuit maximizes at every parameter pair remains a stronger open structural problem.

The argument uses classical ideas—LP/extreme points, signed circuits of vector configurations, determinants, cycle-index recurrences—and extends the orbital primal-dual and all-rank cycle-compression theorems previously documented in this public repository. It makes no unverified priority claim. This is an AI-assisted proof draft, not a proof-assistant kernel formalization or external human referee report. Such independent audits remain important.

9. The $m+2$ support bound is genuinely sharp

Proposition D (minimal support at $(n, k) = (11, 4)$). For the S_{11} action on four-element subsets, the exact optimal coefficient is $C_{11,4} = 1629/4549$. Every signed marginal-preserving perturbation **attaining that optimum** has nonzero mass in at least six distinct short-cycle types. Since $m = 4$, this shows the universal support bound $m+2$ in Theorem A **cannot be reduced to $m+1$ uniformly**.

Proof. The independently certified $n = 11, k = 4$ dual in the repository has coefficient vector

$$\lambda = (-1511/955290, -624/159215, -169/191058, -365/95529)$$

for the four nontrivial orbital counts. The dual function

$$h(g) = \mathbf{1}_{\{g=e\}} - \sum_{j=0}^3 \lambda_j F_j^{(4)}(g)$$

has exact range $[2920/4549, 1]$. An exhaustive integer check over **all 54 feasible short-cycle types** finds that the *only* types attaining its upper value 1 are

$$(11, 0, 0, 0), \quad (7, 0, 0, 1), \quad (0, 1, 3, 0),$$

and the *only* types attaining its lower value $2920/4549$ are

$$(9, 1, 0, 0), \quad (4, 0, 1, 1), \quad (0, 4, 1, 0).$$

For an optimal signed perturbation, the difference between these two dual values must equal the ℓ^1 objective. Equality in the elementary range bound for a zero-mass signed measure forces its positive part to live **entirely on the upper contact set**, and its negative part **entirely on the lower contact set**, after orienting the identity atom positively.

Consider the resulting 5×6 orbital matrix, with identity first and the five other contact types lexicographically ordered. Its exact alternating maximal-minor vector is

$$(-344826720, -23708160, 11430720, 121927680, -594397440, 829573920).$$

Every coordinate is **nonzero**, so the matrix has rank five and a one-dimensional kernel, with every nonzero kernel vector supported on **all six columns**. Its exact ℓ^1 ratio is

$$\frac{2(344826720)}{344826720 + 23708160 + 11430720 + 121927680 + 594397440 + 829573920} = \frac{1629}{4549}.$$

Conjugation averaging cannot increase total variation or change the identity atom, so an arbitrary (not necessarily central) attaining perturbation would yield a central attaining perturbation with no new short-cycle types. It, too, must therefore involve at least six types. This proves sharpness.

Every stated dual contact inequality, complete contact set and integer maximal minor is independently replayed by the published pure-integer/Fraction checker `check_sparse_support_sharpness.py`. The proof is an exact finite certificate, not a numerical LP conclusion. QED.

10. Extension to all finite group actions

The same maximal-minor method works beyond symmetric subset actions, with the cycle-compression-specific rank formula replaced by an explicit rational row-basis selection.

Theorem E (universal finite-group exact-minor formula). Let a finite group G act on a nonempty finite set Ω , without requiring transitivity or faithfulness. Let u_G be uniform on G . A law ν is admissible when its entire image matrix agrees with that of u_G :

$$\nu\{g : g(x) = y\} = u_G\{g : g(x) = y\} \quad (x, y \in \Omega).$$

Let $C(G, \Omega)$ be the optimal single-atom versus TV coefficient, with value zero when no nonuniform admissible law exists.

Let $\mathcal{O}_1, \dots, \mathcal{O}_s$ be the orbitals of G on ordered pairs $\Omega \times \Omega$, and let $K_0 = \{e\}, K_1, \dots, K_{p-1}$ be its conjugacy classes. Define

$$F_j(g) = |\{x \in \Omega : (x, gx) \in \mathcal{O}_j\}|.$$

Build the explicit integer $(s+1) \times p$ matrix with columns

$$a_i = (1, F_1(K_i), \dots, F_s(K_i))^T.$$

Its top row consists of ones, and $\sum_j F_j(g) = |\Omega|$, so the row rank ρ satisfies $1 \leq \rho \leq s$. Choose **any independent set of ρ rows containing the all-ones row**, obtaining a full-row-rank integer matrix $D \in \mathbb{Z}^{\rho \times p}$.

For every ρ -subset $J \subseteq \{1, \dots, p-1\}$, form $(d_0, d_{j_1}, \dots, d_{j_\rho})$ and its alternating $\rho \times \rho$ maximal-minor vector $\Delta(J)$. Then, with the empty maximum defined to be zero,

$$C(G, \Omega) = \max_{\substack{J \subseteq \{1, \dots, p-1\}, |J|=\rho \\ \sum_i |\Delta_i(J)| > 0}} \frac{2|\Delta_0(J)|}{\sum_i |\Delta_i(J)|}.$$

The value is independent of the chosen rational row basis. In particular $C(G, \Omega)$ is **rational**, and, whenever positive, it is attained by an admissible perturbation supported in its signed part on at most $\rho + 1 \leq s + 1$ conjugacy classes.

Proof. Conjugation averaging preserves the distinguished identity atom and the full image-marginal conditions while not increasing total variation. For central laws, equality of each entry of the image transition matrix is equivalent to equality of the s orbital-average counts: the entries are constant on each ordered-pair orbital, and the sum over that orbital is F_j . Therefore every central signed admissible perturbation corresponds exactly to a signed class-mass vector v annihilated by the displayed integer matrix. Passing to the independent rows D does not change this kernel. Conversely a signed vector in $\ker D$ determines central signed masses on the conjugacy classes, which can be added with sufficiently small amplitude to the strictly positive uniform group law. Thus its absolute total variation is exactly one half its class-mass ℓ^1 norm, and

$$C(G, \Omega) = \sup_{v \neq 0, Dv=0} \frac{2|v_0|}{\sum_i |v_i|}.$$

Lemma C, now with row rank ρ , gives precisely (28), its rationality and the support bound. The same maximal-minor vector produces the exact attaining positive and negative class mixtures as in (19). QED.

For the symmetric k -subset action, complementation reduces to $m = \min(k, n - k)$, exact cycle compression replaces every conjugacy class by its short-cycle vector, and Theorem B proves that $\rho = m + 1$. Thus Theorem A is the **explicit, compressed, nondegenerate specialization** of Theorem E, not merely a restatement of finite-group LP duality.

Computational scope: the general group formula requires as input the finite group conjugacy classes and their integer orbital counts, while the symmetric-subset formula computes those columns directly from (n, k) . No uniform elementary branch classification of all finite group actions is asserted.

11. A purely polynomial proof of the binomial rank determinant (preferred for Lean)

Theorem B admits a second, **strictly more elementary** proof that avoids the formal algebraic root $\Lambda(u, t)$, its derivative, and all infinite power-series manipulations. It uses only the exact integer-polynomial trace recurrence (3). This is particularly useful for a finite polynomial-ring Lean formalization.

Retain $n \geq 2m \geq 2$. Let

$$Z_0(u, t) = 2, \quad Z_1(u, t) = 1 + ut, \quad Z_\ell(u, t) = (1 + ut)Z_{\ell-1}(u, t) - u(t-1)Z_{\ell-2}(u, t).$$

For each $0 \leq j < m$, the permutation g_j consists of j fixed vertices and one $(n-j)$ -cycle. Its **exact, untruncated** generating polynomial is

$$G_j(t) = [u^m](1 + ut)^j Z_{n-j}(u, t).$$

Lemma F (exact finite-difference factorization). For every $r, j \geq 0$ with $j + r \leq m - 1$,

$$\boxed{\nabla^r G_j(t) = (t-1)^r [u^{m-r}](1 + ut)^j Z_{n-j-2r}(u, t).}$$

Here $\nabla G_j = G_{j+1} - G_j$. All subscripts in (30) are nonnegative, and the identity takes place in the finite integer polynomial ring $\mathbb{Z}[t]$.

Proof. The trace recurrence rearranges into the exact identity

$$(1 + ut)Z_{L-1} - Z_L = u(t-1)Z_{L-2} \quad (L \geq 2).$$

For $r = 0$, (30) is exactly (29). Suppose (30) holds for r and $j + r + 1 \leq m - 1$. Then

$$\begin{aligned} \nabla^{r+1} G_j(t) &= \nabla^r G_{j+1}(t) - \nabla^r G_j(t) \\ &= [u^{m-r}](t-1)^r (1 + ut)^j [(1 + ut)Z_{n-j-2r-1} - Z_{n-j-2r}] \\ &= [u^{m-r}]u(t-1)^{r+1} (1 + ut)^j Z_{n-j-2r-2} \\ &= (t-1)^{r+1} [u^{m-r-1}](1 + ut)^j Z_{n-j-2(r+1)}, \end{aligned}$$

using (31) with $L = n - j - 2r$. The needed condition $L \geq 2$ follows from $n \geq 2m$ and $j + r + 1 \leq m - 1$. Thus the induction is valid. QED.

In particular, taking $j = 0$ gives

$$\nabla^r G_0(t) = (t-1)^r [u^{m-r}]Z_{n-2r}(u, t).$$

At $t = 1$, the second term in the recurrence for Z_L vanishes. Since $Z_1(u, 1) = 1 + u$,

$$Z_L(u, 1) = (1 + u)^L \quad (L \geq 1).$$

Because $n - 2r \geq 2$, the **first nonzero Taylor coefficient** of (32) at $t = 1$ is therefore

$$\frac{1}{r!} \frac{d^r}{dt^r} \nabla^r G_0(t) \Big|_{t=1} = [u^{m-r}]Z_{n-2r}(u, 1) = \binom{n-2r}{m-r} > 0.$$

The column change $G_j \mapsto \nabla^r G_0$ is unit triangular, and the coefficient change from the monomial basis t^h to $(t-1)^h$ is unit triangular. Equation (32) supplies zero coefficients below the diagonal, and (34) its nonzero diagonal entries. Taking determinants gives

$$\det(F_h(g_j))_{0 \leq h, j < m} = \prod_{r=0}^{m-1} \binom{n-2r}{m-r}.$$

Expanding along the last identity column yields the full statement of Theorem B, with sign $(-1)^m$. **Thus the rank theorem follows entirely from elementary exact polynomial identities in $\mathbb{Z}[u, t]$.**

The independent public checker `check_finite_difference_rank.py` implements (29)–(34) directly with sparse integer-polynomial dictionaries, without importing the maximal-minor algorithm. Its separate test suite covers 30 parameter pairs for $1 \leq m \leq 10$. The **proof above**, not those finite tests, establishes the statement for every n, m .

Lean implication. A formal development can use finitely truncated polynomial arithmetic and two simple recurrences rather than requiring algebraic roots in a formal power-series ring. The full Lean proof is still pending; the existing compiled `KernelMass.lean` certifies only the signed total-mass lemma.